

MODEL QUESTION PAPER – SET-1
SPECIALITY POLYMERS

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	LOI of PS and PPS is different why? Which is high	M 1.01	U
2	Which polymer is used in bullet proof jacket?	M 1.04	A
3	State the name of temperature resistant Polymer used in coffee dispenser, toaster and hair dryer	M 1.03	A
4	What is a conjugated polymer	M 2.01	R
5	What are Photoresists	M 2.03	R
6	PVDF is a ----- polymer a. Photo conducting b. Piezo electric c. Pyro electric d. Electroluminescent	M 3.02	U
7	Which polymers used in xerography	M 2.03	U
8	List out two properties of ionomers	M 3.01	R
9	Chaitin is obtained from-----	M 4.01	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Identify structure of PPO,PES,PEK	M 1.01	U
2	Define LOI. Write any two factors affecting flame resistance.	M 1.012	U
3	List out three conducting polymers with structure	M 2.01	R
4	State piezo electric effect and photo conductivity	M 2.03	R
5	Define OFC .List out the polymers used for (POFC)	M 4.04	U
6	Compare PCC, PMC.	M 4.01	U

7	Compare bio medical and bio degradable polymers	M 4.02	U
8	Differentiate the ionic polymers and poly electrolytes	M 3.01	A
9	Define polymer hydrogel and state two application	M 3.03	U
10	What are the properties and applications of PLA	M 4.02	U

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain structure properties and application of PTFE	M 1.02	U
OR			
2	Explain structure properties and application of PEEK	M 1.02	U
3	Explain the factors affecting thermal stability and four methods to improving the thermal stability of polymers	M 1.01	U
OR			
4	Explain structure properties and application of PPS	M 1.02	U
5	Explain structure properties and application of Poly Acetylene.	M 2.01	U
OR			
6	Explain structure properties and application of PPP	M 2.01	U
7	Explain structure properties and application of Poly Vinyl Carbazole.	M 2.02	U
OR			
8	What is LCP? Explain its properties and application?	M 2.04	U
9	Explain the Properties of ionic polymers- Ion-exchange and Hydrophilicity.	M 3.02	U
OR			
10	Explain the structure properties and application of silicone polymer.	M 3.02	U
11	Explain structure properties and application of nano polymers.	M 4.03	U
OR			
12	Explain the properties and application PHEMA.	M 4.02	U

MODEL QUESTION PAPER – SET-2
SPECIALITY POLYMERS

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Define LOI	MO 1.01	U
2	Which polymer is used in Coffee machines and hair dryers?	MO 1.01	A
3	What are the main properties of PVDF and PVK?	MO 2.01	U
4	Write two applications of photoresist polymers	MO 2.02	U
5	Define polyelectrolyte	MO 2.04	U
6	Nafion and Surlyn are the examples of ----- and----- ionomers	MO 3.03	U
7	Salt water purification under the influence of pressure is known as----- and under electric field is known as --- -----	MO 3.01	A
8	What are the two monomers used in PIC manufacturing	MO 4.02	U
9	List out the two polymers used as the binder in rocket propellants	MO 3.04	U

PART B

V. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	Identify the structure of PPS, PES, PBI	MO 1.01	U
2	What is ladder polymer	MO 1.02	U
3	Write the out structure of PPP, POLY ACETYLENE, and PVDF	MO 2.02	R
4	What are photo resists?	MO 2.01	U

5	What are liquid crystal polymers	MO 2.02	U
6	What are the requirements for propellant polymeric binders?	MO 3.01	U
7	List out the three properties of nano polymers	MO 3.04	U
8	What are the properties and application of Polymeric concrete -PMC	MO4.01	U
9	Define the working principle of Optical fiber cable	MO 4.02	R
10	List out the three applications of ionic polymers	MO 4.04	U

PART C

VI. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain structure properties and application of PTFE	MO 1.02	R
OR			
2	Explain structure properties and application of PPO	MO 1.03	U
3	Explain the four factors affecting thermal stability	MO 1.04	U
OR			
4	Explain four methods to improving the thermal stability and fire resistance of polymers	MO 1.02	R
5	What are conductive polymers, explain the conductive mechanism	MO 2.02	U
OR			
6	Explain Photo conductivity and properties and application of PVK	MO 2.03	R
7	How PVK work in photocopying machine	MO 3.02	U
OR			
8	Explain structure properties and application of Poly Styrene ionomers	MO 3.03	U
9	Explain structure properties and application of TFE ionomers	MO 3.02	U
OR			

10	Explain properties of ionic polymers	MO 3.04	U
11	Explain the preparation properties and application of Polymeric concrete -PIC	MO 4.02	U
OR			
12	Explain preparation properties and application of POFC	MO 4.03	U